EC475 Seminar: Health and Global Development

Colby College Department of Economics

Spring 2022

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Office Hours: Tu 10-11am; W 12:30-2:30pm; Th 9-10:30am; and by appointment
Please sign up for office hours at www.calendly.com/drlafave if you'd like to Zoom

Class: Tuesday 1:00-3:30 Diamond 341 Thursday 1:00-2:15 Diamond 341

Senior Seminar during a Pandemic

The world is (still!) messy right now. The last 24 months have brought to the forefront many acute and simmering realities about justice, inequity, and anxiety. The importance of empathy has been elevated - being kind to ourselves first and extending that same grace toward others. Add in to that that it's now your last semester of college…

You may have lost opportunities you were excited about, endured quarantine, tested positive for COVID-19, and/or been hospitalized. You all have increased responsibilities, reasons to worry about those you care about, and may be facing uncertain post-Colby prospects.

Empathy is my guiding principle this semester. **I'm fully committed to making sure that you learn everything you were hoping to from this class.** I will make whatever accommodations I can to help you finish your work, do well on your projects, and learn and understand the material. Under ordinary conditions, I am flexible when students face difficult challenges. Under pandemic conditions, that flexibility is intensified.

I will not judge you or think less of you if you tell me you're having trouble. I hope you'll extend me the same grace.

You *never* owe me personal information about your health (mental or physical). You are *always* welcome to talk to me about things that you're going through. If I can't help you, I will find somebody who can.

If you need extra help or time, or if you feel behind and that you're not understanding everything, **do not suffer in silence!** Please talk to me. I **promise** I will work with you.

I want you to learn how to process and produce cutting-edge economic research during this class, but I primarily want you to stay healthy, balanced, and grounded during your senior-spring semester.

Course Description

This course exposes students to emerging issues in the economics of global health. By integrating economic theory and recent empirical work using detailed survey data and field experiments, we analyze problems facing developing populations and policies aimed at their solutions. We consider extreme poverty and hunger, child mortality, health care delivery and provider quality, and the relationship between income, poverty, and health. Attention will also

be given to global health policy and empirical evidence on the success and failure of policies that target maternal and infant health, anemia, HIV, malaria, Ebola, and COVID-19. This course heavily relies on applying concepts covered in statistics, econometrics, and intermediate microeconomics to reading, discussing, and conducting empirical research.

Format
We will examine global health through a variety of methods, incorporating in-class analysis with student-led discussions. I will present some background and technical material for the sake of clarity, but the emphasis of this course is placed on participation and intellectual exchange between students.

Pre-requisites
EC224, EC393 and senior standing as an economics major. I expect that you have not only completed, but are comfortable with the material from Intermediate Microeconomics EC223, Research Methods and Statistics EC293, and Econometrics EC393. I know you may have taken Econometrics a while ago or could be currently taking it as well–please reach out if you feel particularly rusty or lost.

Goals of the Course

- Gain an in depth understanding of the successes and ongoing challenges in global health as they relate to the process of economic development.
- Recognize the interdisciplinary nature of social, economic, political, and environmental factors that affect global health.
- Learn how to read and critically evaluate original research. Be able to develop an understanding of the intuition, strengths, and weaknesses behind differing analytical and empirical approaches.
- Improve one's ability to clearly and concisely articulate positions on an issue in written prose, in-class discussions, and oral presentations.
- Connect with the world beyond the classroom, see yourself as a global citizen, and confront challenging issues that may be outside of your comfort zone.

Expectations

Understanding what I expect of you and what you can expect from me is essential for a successful course–not only for doing well in terms of your grade, but also for having an engaging, and dynamic class experience.

What I expect from you:

- I expect you will abide by the College's COVID-19 policy to do your part to keep our campus and local communities safe.
- I expect you will uphold the Colby Affirmation and demonstrate academic honesty in all parts of the course. Students found to be in violation will be referred to the appropriate Dean. Colby takes these issues very seriously.
- Maintain an active engagement in the course. Pay attention, and respond when called upon. Be engaged outside of class as well; follow along with the course content, prepare and complete assignments by their due dates.
- Come to class prepared: ****It is <u>essential</u> that you come having looked at the posted material.**** The course moves at a pace that reflects this expectation.

- Help guarantee that we create an inclusive environment where all students are well-served by this course, and one that recognizes the strength and benefit of diverse perspectives.
- Actively participate in the learning process. Help each other to establish a collaborative learning environment. Ask questions when you don't understand something. Come to office hours if you're confused – use the available resources to set yourself up for success

What you can expect from me:

- I will follow all aspects of the College's COVID-19 policy.
- I will come to class prepared, use class time effectively, and foster a learning environment that is engaging, inclusive, and respectful. I will gladly honor your request to address you by an alternate name and/or gender pronoun than what is listed on the official class roster– please advise me of this on the online Day 1 survey.
- I will present economics so that it is relevant to your lives, not only a set of abstract concepts. I will work with you to build knowledge and develop a new lens through which to view the world.
- Assess your learning fairly and provide constructive feedback on all presentations and assignments.
- Be available and approachable. I will be available for office hours before class to work with discussion leaders, and work with those who have time conflicts to find appointments that work for both of us. I will do my best to respond to all emails with ec475 in the subject line within 12 hours.

Grading

The majority of work in the course will be staying on top of the reading on a weekly basis and continued progress on your research project. This is no small task–our material is drawn from leading academic journals, and we will cover a number of papers each class. Your active, respectful participation in class discussions is expected.

- **Participation:** A seminar course is only successful if we are all equally invested in participating and getting the most out of the experience. Your participation grade will reflect your attendance, contribution to discussions, and demonstrated knowledge of the readings. This means more than just a surface reading of the assigned material–be prepared to challenge assumptions, suggest extensions, and go beyond what you've read in the text. I encourage you to look into readings, cases, and other materials that complement the assigned material.

  Disruptive, dismissive, inappropriate, or disrespectful comments or actions have no place in a seminar. Please refrain from texting, tweeting, emailing, snap-chatting, instagram-ing, posting, etc. during class. Our class is a place to share ideas and views in an environment that's comfortable for asking questions and contributing to discussions.

- **Problem Set:** You have one single (significant) problem set due at the end of week four that combines some of the foundational theory from the first three weeks of class with empirical work using Stata (or R or python if you prefer).

- **Discussion Leadership:** You will be responsible for leading a portion of class on several occasions on one of the assigned articles of the week.

- The first discussions are to be done in pairs; the next set are solo affairs.
- Each discussion should last approximately 30 minutes.
- I have posted a handout about discussing scholarly work, and I will model how these experiences should proceed during the first few weeks.
- I will be available prior to our Tuesday classes for last-hour questions from discussion leaders.

- **Discussion Questions:** You have a responsibility to the group to ask and answer questions, and push the discussion to a deep level. To help in guiding the discussion leader(s), you must submit **two (2) questions for each required reading** by **Monday at 9pm** through **the Google Forms linked on Moodle**. The discussion leaders are then expected to use these questions to help guide an active discussion concerning the work they are responsible for the next day. The depth and quality of your questions reflects your acknowledgement of a **shared responsibility** in the course.

- **Research Project:** As the capstone course in the major, you are all equipped with the tools and background to conduct creative, original empirical research. The largest part of your individual grade is a capstone research project. Further instructions will follow throughout the term, but the major parts are as follows:

  - **Research Proposals:** Given the challenges in finding a clearly defined question that can be answered with available data, the first component of the project is a proposal of **two (2)** possible research questions related to global health. For each topic you'll need to discuss why the research question is interesting and important, what your hypothesis is, how you intend to test it, and what data you will use. I'll provide extensive feedback on each topic in a one-on-one meeting and you will choose one topic to develop into an empirical research paper.

  - **Data Summary:** Preparing data can be very time consuming–almost always more time consuming than you initially expect. The data summary assignment lets each of us know that you have the data in a usable form in time to complete the final paper and allows us to spot potential pitfalls.

  - **Working Paper:** A substantially complete draft due mid-way through the semester so that I can provide feedback on your progress.

  - **Peer review:** The skills you develop reading and discussing articles for class are not only relevant for assessing work produced by professional academics – you will be responsible for reading and providing comments on a classmate's working paper as well.

  - **Presentation:** Approximately 10-minute discussion of your work in progress during the last weeks of class.

  - **Final Paper:** Not to exceed 15 pages, the final draft should incorporate feedback from your working paper, peer review, and presentation.

- Your final grade will be determined according to the following weights:

  Problem set: 5%
  Participation: 15%
  Discussion questions: 10%
  Discussion leadership: 20%
  8% for group papers
  12% individual papers
  Research Project: 50%

5% proposal	5% data summary	10% working paper
5% peer review	10% presentation	15% final

- *A Note on Re-grade Requests:* You may resubmit any assignment with an additional sheet attached explaining in a few words why you deserve a better grade. These are due one week after the graded work is returned. If you choose to petition for a change to your grade, I will re-grade the **entire** work - not only the part in question.

Key Dates (subject to change)

Feb 3 (Th)	Day #1
Feb 25 (F)	Problem set due
Mar 7 (M)	Research proposals due
Mar 8, 10 (Tu-Th)	Individual research meetings replace class
Mar 22-25	Spring Break
Apr 1 (F)	Data summary due
Apr 15 (F)	Working paper due
Apr 21 (Th)	Peer review due
Apr 28-29 (Th, Fr)	No classes – College mental health day
May 3, 5 (Tu, Th)	In-class research presentations
May 6 (F)	Final paper due

Texts and Readings

Course readings are drawn from leading economic and public health journals. Texts listed below provide additional background information to supplement the articles used in class or for your research project. Colby has access to electronic versions of the majority of these or you can borrow them from my office.

- Jacobsen *Introduction to Global Health*
- Levine (ed.) *Case Studies in Global Health*
- Merson, Black and Mills *Global Health*
- Jones, Rice, d'Uva and Balia *Applied Health Economics*
- Banerjee and Duflo *Poor Economics*
- Ray *Development Economics*
- Banerjee, Benabou, and Mookherjee (eds.) *Understanding Poverty*

Course Outline *(tentative and subject to change)*

Week 1. Intros and What is Global Health (Feb 3)

What role does health play in improving the well-being of individuals, communities, and nations?

Optional readings:

Koplan, J. P. et al. "Towards a common definition of Global Health," *The Lancet*, 373 (9679), 2009.

Optional

Strauss, John and Duncan Thomas "Health over the life course" in T. P. Schultz and J. Strauss, eds, *Handbook of Development Economics*, vol. 4, 2007.

Week 2. The Role of Economics in Global Health (Feb 8, 10)

What is the state of population health around the world? What tools do economist use to answer questions in global health? How can economic theory inform a study of health?

Readings:

Dupas, Pascaline "Health Behavior in Developing Countries," *Annual Review of Economics*, vol. 3, 2011.

Roser, Max, "Stop saying 2016 was the worst year ever," *Washington Post.*

Data Visualization Slide Assignment

Week 3. Empirical Approaches to Global Health (Feb 15, 17)

How do we measure health? What is the causal relationship between health and wealth? Is there "low hanging fruit," available to improve health? How would you find out?

Readings

Burke, M., E. Gong, and K. Jones, "Income Shocks and HIV in Africa," *Economic Journal*, 125 (585): 1157-1189, 2015.

Miguel, T. and M. Kremer, "Worms: Identifying impacts on education and health in the presence of treatment externalities," *Econometrica*, 72(1): 159-217, 2004.

Week 4. Behavioral Insights in the Supply of Healthcare and Health Workers (Feb 22, 24)

What is the state of health service delivery in developing areas? What are the primary barriers to quality care? How can we best design policies to improve the quality and quantity of care?

Readings

Banerjee, Abhijit, Jishnu Das, Jeffrey Hammer, Reshmaan Hussam, Aakash Mohpal , "The Market for Healthcare in Low Income Countries," *Working Paper,* December 2020.

Ashraf, N., O. Bandiera, and K. Jack. "No margin, no mission? A field experiment on incentives for pro-social tasks." *Journal of Public Economics*, 120: 1-17, 2014.

Ashraf, N., O. Bandiera, and S. Lee, "Losing Prosociality in the Quest for Talent? Sorting, Selection, and Productivity in the delivery of Public Services." *American Economic Review*, 2020.
Policy Video #1 describes the project in 8 min detail
Policy Video #2 summarizes the study in 2 min

Optional

Abarcar, P., C. Theoharides, "Medical Worker Migration and Origin-Country Human Capital: Evidence from U.S. Visa Policy," *Review of Economics and Statistics*, 2021.

Banerjee, A., E. Duflo, and R. Glennerster, "Putting a Band-Aid on a corpse: Incentives for nurses in the Indian public health care system" *Journal of the European Economic Association*, 6(2): 487-500, 2008.

Das, J., J. Hammer, and Leonard, K.L., "The Quality of Medical Advice in Low Income Countries," *Journal of Economic Perspectives,* 22 (2), 2008.

Basinga, P., P. Gertler, A. Binagwaho, A. Soucat, J. Sturdy, and C. Vermeersch, "Effect on maternal and child health services in Rwanda of payment to primary health-care providers for performance: an impact evaluation," *Lancet*, 377: 1421-28, 2011.

Leonard, K.L. and M.C. Masatu, "Using the Hawthorne Effect to examine the gap between a doctors best possible practice and actual practice," *Journal of Development Economics*. 93 (2): 226-243, 2010.

Svensson, J. and M. Bjorkman "Power to the People: Evidence from a Randomized Field Experiment of Community-Based Monitoring in Uganda," *Quarterly Journal of Economics*, 124:2, 2009.

Das, J., Chowdhury, A., Hussam, R., & Banerjee, A. V. "The impact of training informal health care providers in India: A randomized controlled trial," *Science*, 354(6308), 2016.

Week 5. Free-disbursement vs. cost sharing: How should we price health? (Mar 1, 3)

Is price the only barrier to adoption of health products? Does demand ever slope upward? Can we have both sustainable and successful health campaigns?

Readings

Ashraf, N. J. Berry, and J. M. Shapiro. "Can higher prices stimulate product use? Evidence from a field experiment in Zambia." *American Economic Review*, 100(5): 2383–2413, 2010.

Cohen, J. and P. Dupas. "Free distribution or cost-sharing? Evidence from a randomized malaria prevention experiment." *Quarterly Journal of Economics*, 125, 2010.

Meredith, Jennifer, Jonathan Robinson, Sarah Walker, and Bruce Wydick, "Keeping the doctor away: Experimental evidence on investment in preventative health products," *Journal of Development Economics,* 105, 2013.

Optional

Cohen, J., P. Dupas, and S. Schaner. "Price subsidies, diagnostic tests, and targeting of malaria treatment: Evidence from a randomized controlled trial." *American Economic Review*, 2015.

Kremer, M. and E. Miguel. "The illusion of sustainability." *Quarterly Journal of Economics*, 122(3): 1007– 1065, 2007.

Sarriot, E., E. Swedberg, and J. Ricca. "Pro-sustainability choices and child deaths averted: from project experience to investment strategy." *Health Policy and Planning*, 26(3), 2011.

Tarozzi, A., A. Mahajan, B. Blackburn, D. Kopf, L. Krishnan, and J. Yoong. "Micro-loans, insecticide-treated bednets and malaria: Evidence from a randomized controlled trial in Orissa (India)." *American Economic Review*, 2014.

Week 6. Research meetings replace regular class (Mar 8, 10)

Week 7. Sustainable Development, Natural Environment, and Health (Mar 15, 17)

Why is clean-water an important investment? Can there be health benefit from consequences to improving community infrastructure? What are the benefits of natural capital? What policies are effective in tackling pollution, and how do we measure them?

Readings

Buchmann, Nina, Erica Field, Rachel Glennerster, and Reshmaan Hussam. "Throwing the baby out with the drinking water: Unintended consequences of arsenic mitigation efforts in Bangladesh." Working Paper, April 2019.

Berazneva, J. and T. Byker, "Impacts of Environmental Degradation: Forest Loss, Malaria, and Child Outcomes in Nigeria," Working Paper, 2021.

Barwick, P., S. Li, L. Lin, and E. Zou. "From Fog to Smog: The Value of Pollution Information." Working paper, 2020.

Optional

Greenstone, Michael, Guojun He, Shanjun Li, and Eric Zou, "China's War on Pollution: Evidence from the first five years," Working Paper, 2021.

He, G., M. Fan, M. Zhou. "The effect of air pollution on mortality in China: Evidence from the 2008 Beijing Olympic Games." *Journal of Environmental Economics and Management*, 79, 2016.

Rosales-Rueda, M. "The impact of early life shocks on human capital formation:

evidence from El Niño floods in Ecuador," *Journal of Health Economics*, 62, 2018.

LaFave, D., S. Dissanayake, et al. "Health Impacts of Improved Biomass Cookstoves: Three-year Evidence from Rural Ethiopia," *World Development,* 141, 2021.

Burke, M., S. Hsiang, and E. Miguel, "Climate and Conflict," *Annual Review of Economics*, 7, 2015.

Deschenes, O., "Temperature, human health, and adaptation: A review of the empirical literature," *Energy Economics*, 46, 2014.

Garg, Teevrat, "Ecosystems and Human Health: The Local Benefits of Forest Cover in Indonesia." *Journal of Environmental Economics and Management,* 98, 2019.

Kremer, M. E. Miguel, S. Mullainathan, C. Null, and A. P. Zwane. "Social engineering: Evidence from a suite of take-up experiments in Kenya." Working Paper, April 2011.

Kremer, M., J. Leino, E. Miguel, and A. P. Zwane. "Spring cleaning: Rural water impacts, valuation, and property rights institutions." *The Quarterly Journal of Economics*, 126(1):145–205, 2011.

Jeuland, M., S. Ozdemir; D. Fuente; M. Allaire; D. Whittington. "The long-term dynamics of health benefits from improved water and sanitation in developing countries." *PLoS One,* 8.10, 2013.

Keskin, P., G. K. Shastry, and H. Willis. "Water Quality Awareness and Breastfeeding: Evidence of Health Behavior Change in Bangladesh," *Review of Economics and Statistics,* Forthcoming.

Quick, R. E., Kimura, A., Thevos, A., Tembo, M., Shamputa, I., Hutwagner, L., and Mintz, E. "Diarrhea prevention through household-level water disinfection and safe storage in Zambia." *American Journal of Tropical Medicine and Hygiene*, *66*(5), 584-589, 2002.

Cutler, D. and G. Miller. "The role of public health improvements in health advances: The twentieth-century United States." *Demography*, 42(1):1–22, 2005.

Graff Zivin, J. and M. Neidell. "Environment, Health, and Human Capital," *Journal of Economic Literature*, 51(3): 689-710, 2013.

Week 8. Structural Inequality, Race, and Health in the US (Mar 29, 31)

How does race and racism impact health outcomes? Does diversity matter for health? What are the long-run consequences of early life (dis)advantage? What can policy do to improve inequality and inequity?

Readings

Williams, David, Jourdyn Lawrence, and Bridgette Davis, "Racism and Health: Evidence and Needed Research," *Annual Review of Public Health,* 40, 2019.

Alsan, Marcella and Marianne Wannamaker, "Tuskegee and the Health of Black Men," *Quarterly Journal of Economics,* 133(1), 2018.

Alsan, Marcella, Owen Garrick, and Grant Graziani, "Does Diversity Matter for Health? Experimental Evidence from Oakland," *American Economic Review,* 109(12), 2019.

Optional

Chen, Irene Y., Shalmali Joshi, and Marzyeh Ghassemi. "Treating health disparities with artificial intelligence." *Nature medicine* 26.1 (2020): 16-17.

Evans, Michele K., Rosenbaum, L., Malina, D., Morrissey, S., & Rubin, E. J. "Diagnosing and treating systemic racism." *New England Journal of Medicine,* 383(3), 2020.

Ganong, Peter, Damon Jones et al. "Wealth, Race, and Consumption Smoothing of Typical Income Shocks," Working Paper, 2020.

Obermeyer, Ziad, Brian Powers, Christine Vogeli, and Sendhil Mullainathan, "Dissecting racial bias in an algorithm used to manage the health of populations," *Science*, 266, 2019.

Washington, Harriet, *Medical apartheid: The dark history of medical experimentation on Black Americans from colonial times to the present*, Doubleday, 2006.

Week 9. Legacy of colonization in global health (Apr 5, 7)

Readings

Anderson, Siwan, "Legal Origins and Female HIV," *American Economic Review,* 108 (6), 2018.

Lowes, Sara and Eduardo Montero, "The Legacy of Colonial Medicine in Central Africa," *American Economic Review,* 111(4), 2021.

Archibong, Belinda and Francis Annan, "We are not guinea pigs: The effects of negative news on vaccine compliance," Working Paper, 2021.
2000 *Washington Post* piece

Optional

Huillery, Elise, "History Matters: The Long-Term Impact of Colonial Public Investments in French West Africa." *American Economic Journal: Applied Economics,* 1 (2): 176–215, 2009.

Cagé, Julia, and Valerie Rueda. "Sex and the Mission: The Conflicting Effects of Early Christian Missions on HIV in Sub-Saharan Africa." *Journal of Demographic Economics* 86 (3), 2020.

Nunn, N. "The Long-Term Effects of Africa's Slave Trades." *Quarterly Journal of Economics* 123 (1), 2008.

Michalopoulos, Stelios and Elias Papaioannou, "Pre-Colonial Ethnic Institutions and Contemporary African Development." *Econometrica* 81 (1): 113–52, 2013.

Michalopoulos, Stelios and Elias Papaioannou, "National Institutions and Subnational Development in Africa." *Quarterly Journal of Economics* 129 (1): 151–213, 2014.

Michalopoulos, Stelios and Elias Papaioannou "The Long-Run Effects of the Scramble for Africa." *American Economic Review* 106 (7): 1802–48, 2016.

Week 10. Global mental health (Apr 12, 14)

Readings

Ridley, M. et al., "Poverty, Depression, and Anxiety: Causal Evidence and Mechanisms," *Science* 370, 1289, 2020.

Adhvaryu, A., J. Fenske, and A. Nyshadham, "Early life circumstance and adult mental health," *Journal of Political Economy*, 127(4), 2019.

Blattman, C. et al., "Reducing Crime and Violence: Experimental Evidence from Cognitive Behavioral Therapy in Liberia," *American Economic* Review, 107(4), 2017.

Optional

Haushofer, J., M. Chemin, C. Jang, and J. Abraham, "Peace of Mind: Health Insurance Reduces Stress and Cortisol Levels – Evidence from a Randomized Experiment in Kenya," *Journal of Development Economics,* 2020.

Week 11. Opioids (Apr 19, 21)

Background listening: *Hidden Brain* – Life, Death, and the Lazarus Drug: Confronting America's Opioid Crisis

Readings

Alpert, A., W. Evans, E. Lieber, and D. Powell, "Origins of the Opioid Crisis and its Enduring Impacts," *Quarterly Journal of Economics*, 2022.

Barrette, E., L. Dafny, and K. Shen, "Do Policies to Increase Access to Treatment for Opioid Use Disorder Work?," *NBER Working Paper,* 2021.

Doleac, J. and A. Mukherjee, "The Effect of Naloxone Access Laws on Opioid Abuse, Mortality, and Crime." *Journal of Law and Economics*, 2021.

Optional

Alpert, A., D. Powell, and R. Liccardo Pacula, "Supply-Side Drug Policy in the Presence of Substitutes: Evidence from the Introduction of Abuse-Deterrent Opioids," *American Economic Journal: Economic Policy*, 10(4), 2018.

Abouk, R., R. Liccardo Pacula, D. Poweel, "Association Between State Laws Facilitating Pharmacy Distribution of Naloxone and Risk of Fatal Overdose," *JAMA Internal Medicine*, 2019.

Bao, Yuhua, Yijun Pan, Aryn Taylor, Sharmini Radakrishnan, Feijun Luo, Harold Alan Pincus, and Bruce R. Schackman. "Prescription Drug Monitoring Programs Are Associated with Sustained Reductions in Opioid Prescribing by Physicians." *Health Affairs,* 35, 2016.

Deiana, C. and L. Giua, "The US Opioid Epidemic: State Laws, Prescription Opioids and Crime," Working Paper, August, 2019.

Evan, B., E. Lieber and P. Power, "How the Reformulation of Oxycontin ignited the heroin epidemic." *Review of Economics and Statistics,* 101(1): 1-15, 2019.

Rees, D., J. Sabia, L. Argys, D. Dave, J. Latshaw, "With a Little Help from My Friends: The Effects of Good Samaritan and Naloxone Access Laws on Opioid-Related Deaths," *Journal of Law and Economics*, 62, 2019

Marthur, N. and C. Ruhm, "Marijuana Legalization and Opioid Deaths," Working paper, 2022.

Week 12. COVID-19; impact, policy, and vaccines (Apr 26, 28)

Readings

Egger, D. et al., "Falling living standards during the COVID-19 crisis: Quantitative evidence from nine developing countries," *Science Advances*, 7 (6), February, 2021.

Miguel, E. and A. M. Mobarak, "Economics of the COVID-19 Pandemic in Poor Countries," Working Paper, October, 2021.

Abaluck, J. et al. "Impact of community masking on COVID-19: A cluster-randomized trial in Bangladesh," *Science*, 375 (160), January, 2022.

Optional

Camilo Castro, J. et al. "Market design to accelerate COVID-19 vaccine supply," *Science*, February, 2021. **note: includes the technical appendix**
Athey and Kremer discussion re: advanced market vaccine commitments

Banerjee, A. A Chandrasekhar, E. Duflo, and M. Jackson, "Using Gossips to Spread Information: Theory and Evidence from Two Randomized Controlled Trials," *Review of Economic Studies*, 2019.